Software Engineering

Course 1 – 24 February 2025

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Types of Requirements from Clients

- *Build an application for a hospital. The application will allow you to manage patients (with details about them), manage doctors (with the possibility of scheduling in the intervals allocated to patient consultations), manage nurses (with managing the schedule when they have to come to work, the guards they have to do overnight) , etc. For each patient, the diseases they suffer from, the treatments they must follow, the funds available from the insurance company, etc. will be known.*

- How should we approach such a request?

Context

- **Large applications** (millions of lines of code) written over several months, years ...
- **Working teams**: Project managers, analysts, architects, programmers, testers, support engineers (the order of 10, 100, 1,000 people ...)
- The solution to these problems is a solution written in an **object-oriented programming language**

Motivation 1

- More and more systems are controlled by software: traffic control (air, rail, road, water, etc.), banks, mobile telephony, Internet, supermarket, smart homes, AI
- Economies of all states depend on software
- Software Engineering proposes theories, methodologies and tools for **professional software development**

Motivation 2

- **1946** Goldstine and von Neumann: "1000 guidelines are a reasonable upper limit to the complexity of issues which can be designed as done by computer"

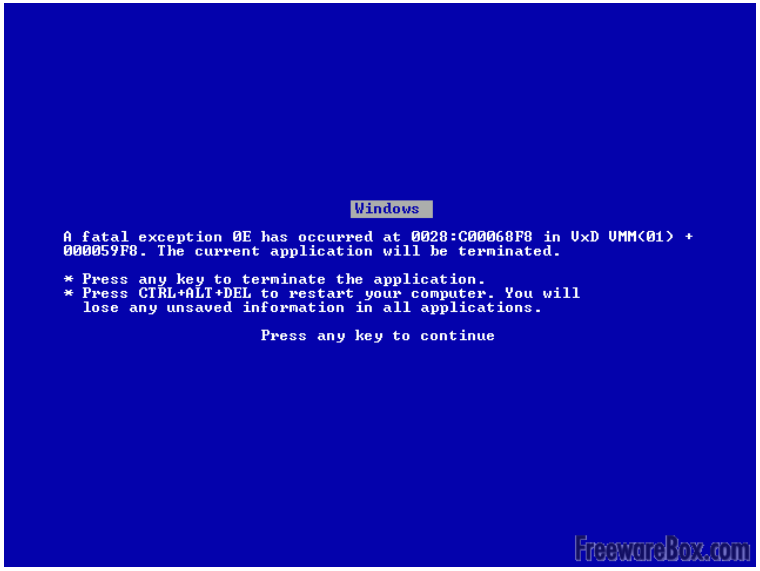
- Ticket booking system for airline KLM contained, in **1992**, **two million lines of code** in assembly language

Motivation 3

- System V operating system version 4.0 (UNIX) was obtained by compiling the 3.7 million lines of code
- Programs written for the NASA space shuttle had around 40 million lines of code object
- To create the IBM OS360 operating system 5,000 man-years were required

Motivation – Famous errors 1

- Huge electricity bills for pensioners, recalculation of pensions



- IBM OS360 contained in each relaunching 1,000 mistakes. *Resignation*…

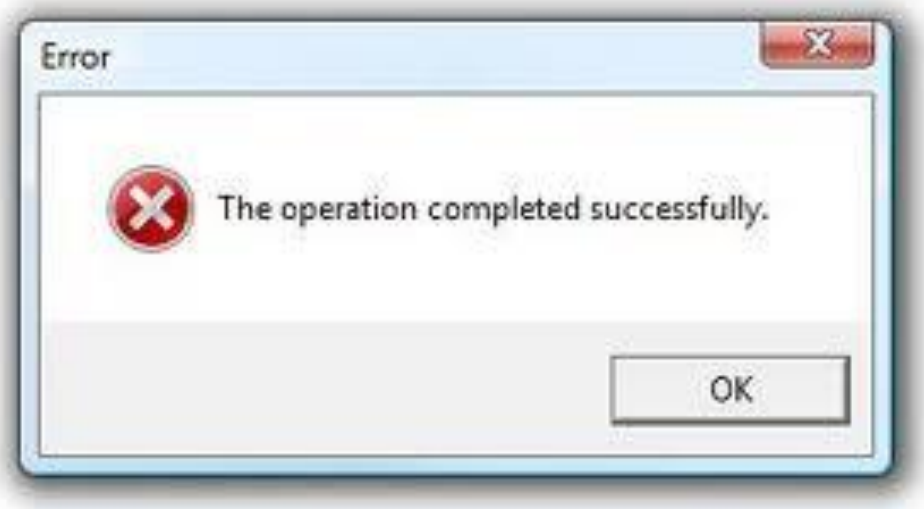
Motivation – Famous errors 2

- *Unlucky programmer at a bank:* Bank wanted to send a special letter to important customers by mail to notify them various services
- The programmer wrote a program that selected 2,000 customers and wrote a personalized letter
- In testing he used a pseudonym client **Rich Bastard**
- Sadly 2,000 customers have received a letter that began "**Dear Rich Bastard** ..."

Motivation – Famous errors 3

- Sacramento: dentist receives in the mailbox 16,000 forms for payment of fees in one week – *"It Was a computer problem"* an official said

- *"Failure to convert English measures to metric values was the root cause of the loss of the Mars Climate Orbiter…"*



Motivation – Famous errors 4

- Venus exploration vehicle loss. Ah, it was actually **.,.** in FOR! ...
- Missile warning system activated. **Attack** or not?
- Ariane 5 explodes. Fireworks Cost: **$ 500 million**

Motivation – Statistics 1

- These are not just "funny errors"

- Large projects may be the most complicated products made by anyone
- Let's look at the statistics
  - We consider that a software project is **successful** if it is done in a reasonable period of time and with a reasonable budget
  - **A failure** of a software product occurs when the product is not achieved or it can not be used

Motivation – Statistics 2

- **Successful studies: USA'82 – Gibson & Singer**
- **18 projects**
  - Success: 17%
  - Partially in use: 28%
  - Satisfactory 11%
  - Failure: 22%
  - Unrated: 11%
- **Reasons for failures**
  - Organizational problems
  - The new working methods and salary policies
  - Unforeseen changes in business

Motivation – Statistics 3

- Successful studies – ONNI'88 (Finland)

- From over 100 projects
  - Success: 33%
  - Problem: 42%
  - Failure: 25%

- Reasons for failures
  - Poor training of software engineers
  - Insufficient resources
  - Management issues

Motivation – Statistics 4

- **The Robbins-Gioia Survey (2001),** Alexandria – Virginia, made a study over the perception by enterprises of their implementation of an E.R.P. (Enterprise Resource Planning) package.
- 232 survey respondents (36 % had, or were in the process of, implementing an ERP system)
- **51 % viewed their ERP implementation as unsuccessful**,
- **46 % did not understand how to use the system**
- 56 % of survey respondents noted that their organization has a program management office (PMO) in place, and of these respondents, only 36 % felt their ERP implementation was unsuccessful

Motivation – Statistics 5

- **The Conference Board Survey (2001)**
- Of 117 companies that attempted ERP implementations
- **34 % were very "satisfied"**
- **58 % were "somewhat satisfied"**
- **8 % were unhappy** with what they got.
- **40 % of the projects failed** to achieve their business case within one year of going live

- The companies that did achieve benefits said that achievement took **six months longer** than expected.
- Implementation costs were found to average 25 % over budget

PE – Definition 1

- First definition (NATO,1968): *the establishment and use of sound engineering principles in order to economically obtain software that is reliable and works efficiently on real machines*

- IEEE Definition (*IEEE Standard Glossary of Software Engineering Technology, 1983): The application of a systematic, disciplined, quantifiable approach to the development, operation, and maintenance of software*

PE – Definition 2

- Refers to two things:
  - "Software engineer" who replaced "programmer"
  - "Software Engineering" is used to describe "building of software systems which are so large or so complex that they are built by a team or by teams of engineers“ – *Fundamentals of Software Engineering* (Ghezzi, Jazayeri, and Mandrioli)

PE – Definition 3

- **FreeDictionary**: "The process of **manufacturing** software systems. A software system consists of executable computer code and the supporting documents needed to manufacture, use, and **maintain** the code."
- **Webopedia.com**: "The computer science discipline concerned with **developing large applications**. Software engineering covers not only the technical aspects of building software systems, but also **management issues**, such as **directing programming teams**, **scheduling**, and **budgeting**. "

PE – Definition 4

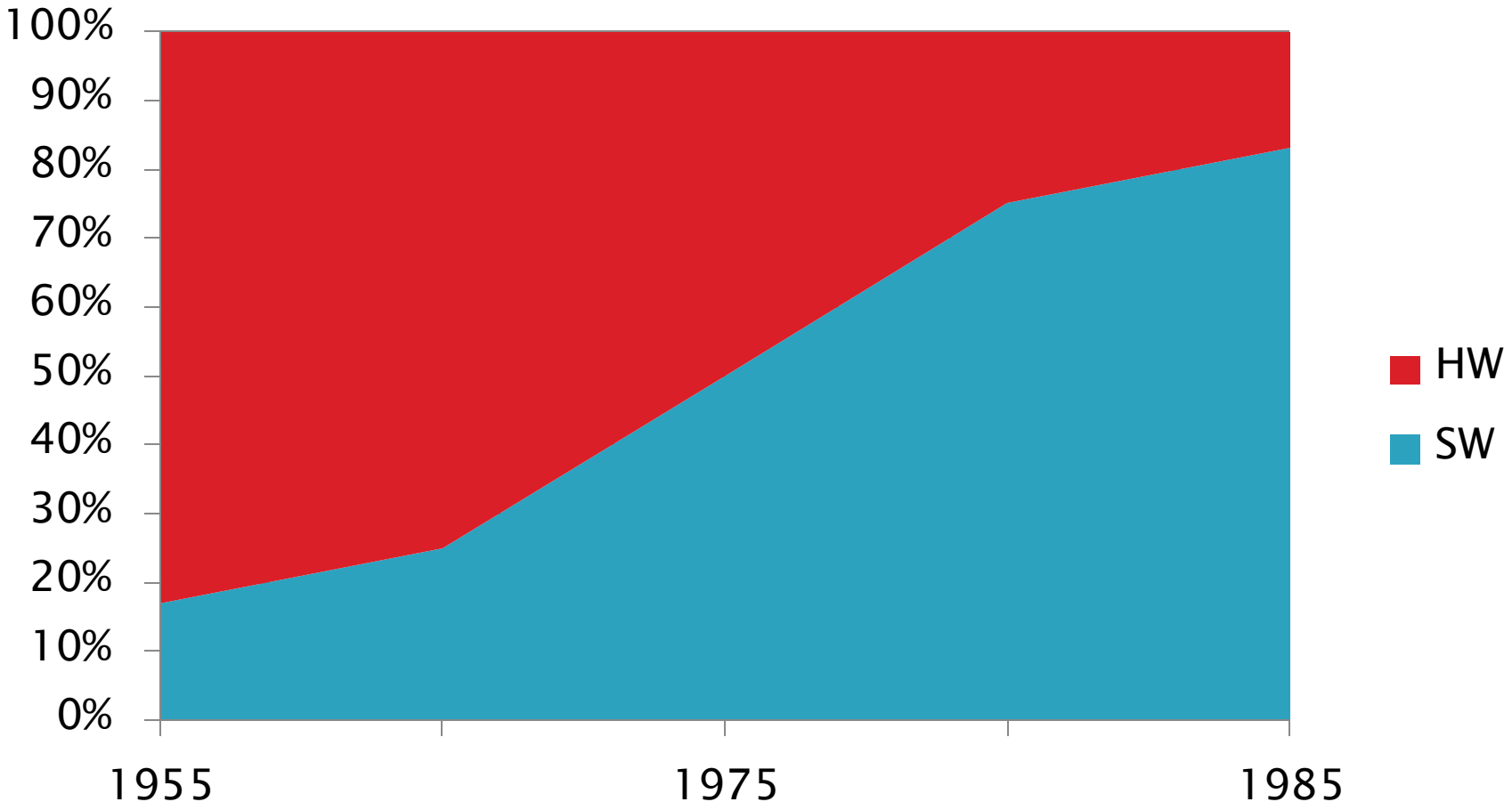
- **Wikipedia**: "**Software engineering** is the application of a systematic, disciplined, quantifiable approach to the **development**, operation, and **maintenance** of software, and the study of these approaches"

- **Answers.com**: "The **systematic** application of scientific and technological knowledge, through the medium of sound engineering principles, to the production of computer programs, and to the **requirements definition**, **functional specification**, **design description**, **program implementation**, and **test methods** that lead up to this code"

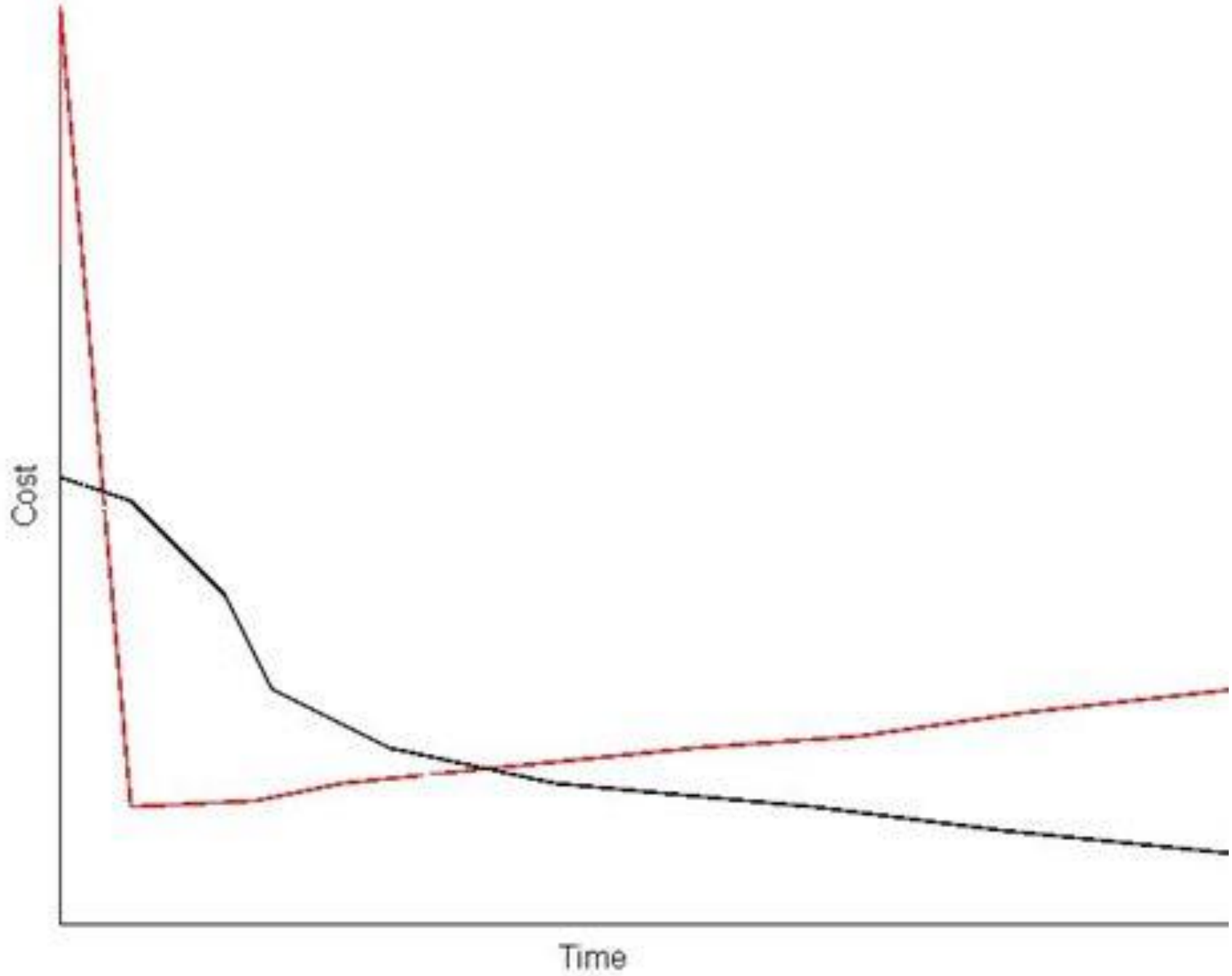
PE – In short

- It is an engineering discipline that deals with all the aspects of developing a large program, by a team of developers
- Proposes a systematic and organized approach to the software development process
- It proposes the use of appropriate techniques and tools taking into consideration:
  - The problem to be solved
  - restrictions
  - available resources

Software vs Hardware costs



Software build (black) vs. buy (red)



Software vs Hardware costs (2)

- Currently the costs for software are higher than the costs of hardware components
- The cost of maintaining a program is usually higher than the cost of creating a program
- Software – represent the programs and the associated documentation

Attributes of a good program

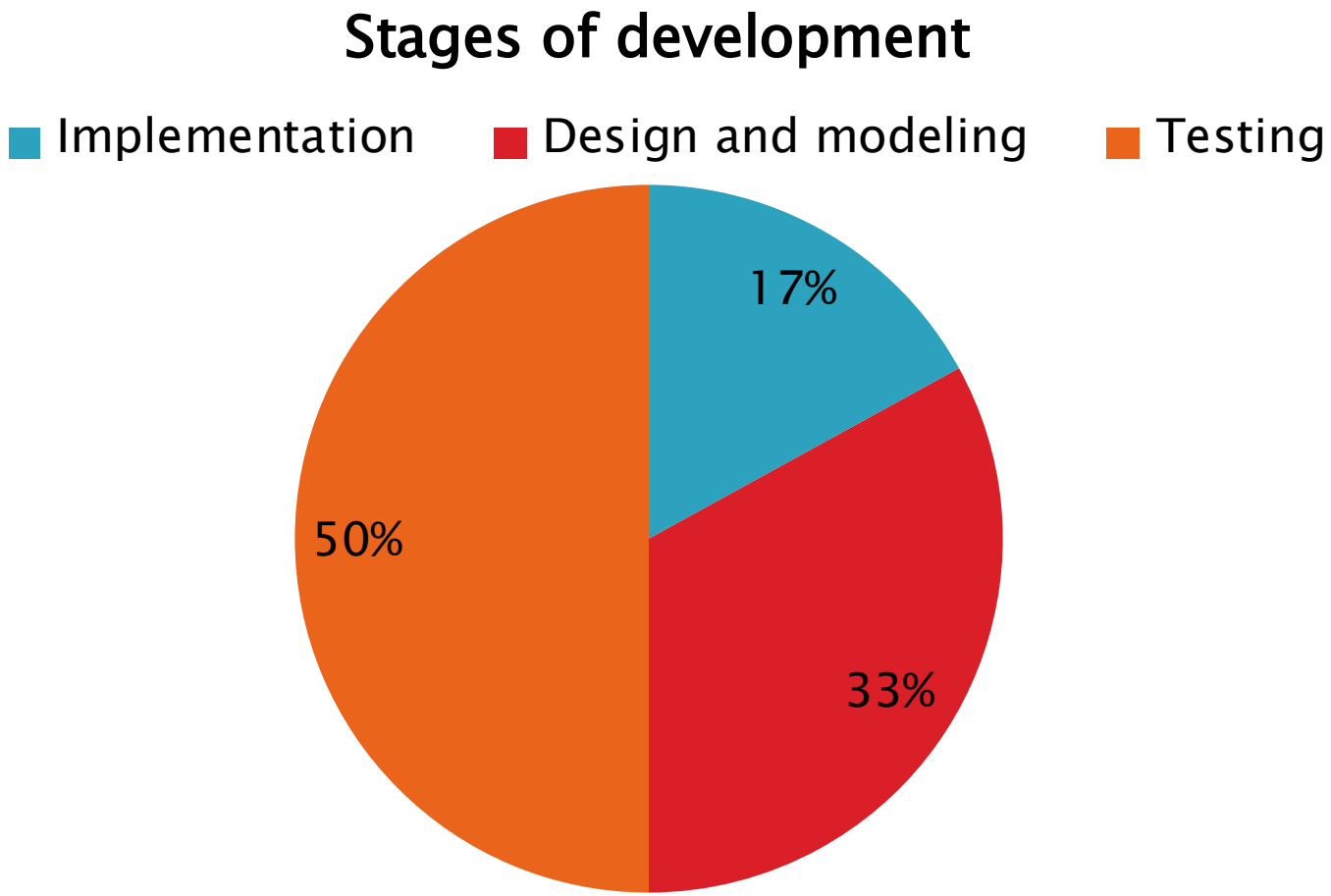
- To provide the requested functionality
- Be easily modified, extended, changed
- Be safe
- Not wasting hardware resources
- Easy to use

Comparison of SE

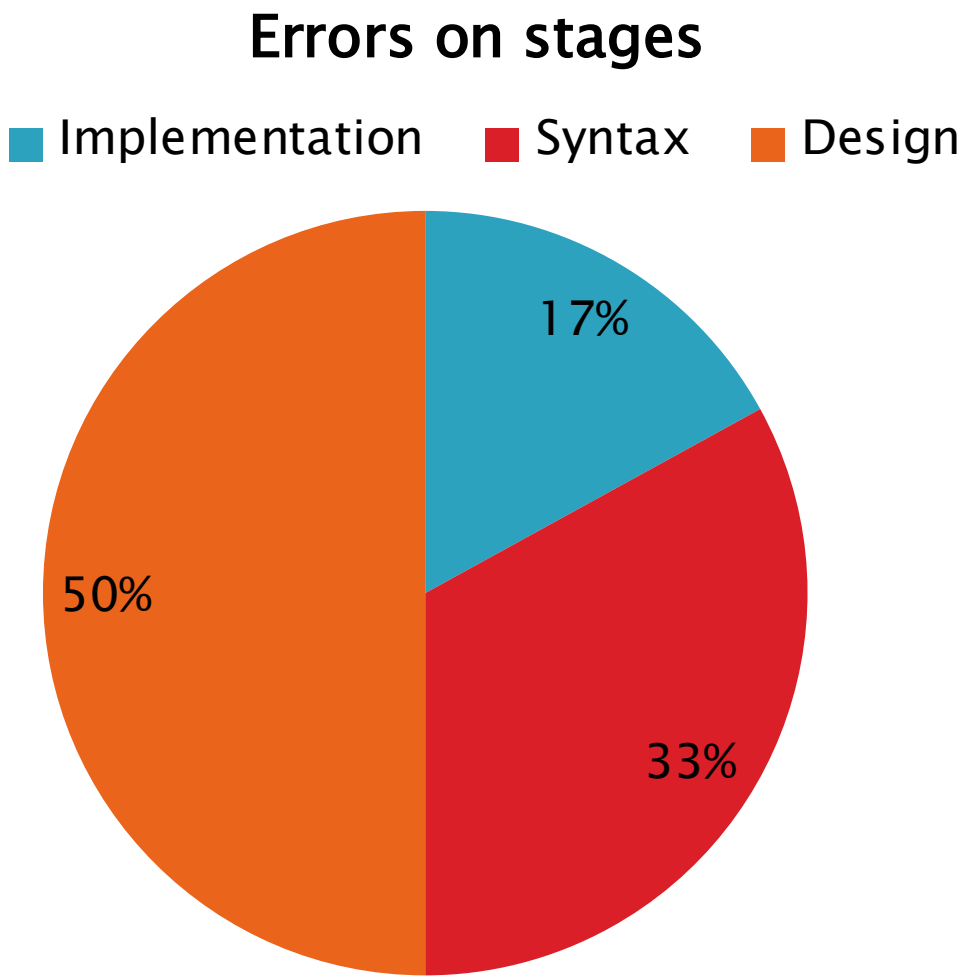
- SE vs computer science
  - Computer science deals with theoretical aspects of software development
  - SE deals with practical aspects of software development

- SE vs systems engineering
  - Systems engineering deals with all aspects of development of computer systems (hardware, software, process engineering)
  - SE is part of systems engineering and is responsible for:
    - Requirements Engineering
    - Architectural and detailed design
    - Implementation
    - Testing
    - Integration
    - Deployment

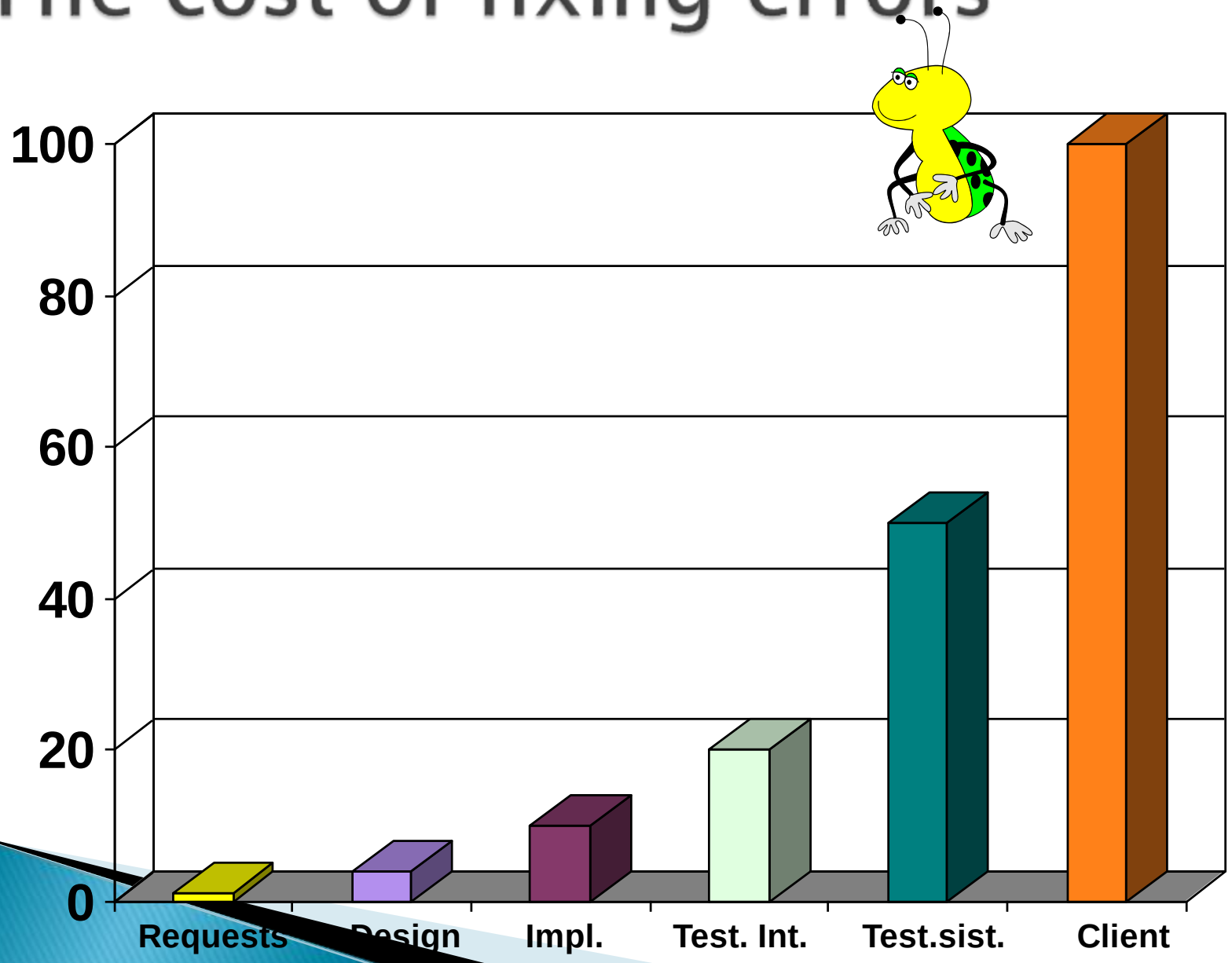
Distribution of costs on stages



Distribution of errors on stages



The cost of fixing errors



Difficulties in SE

- Older systems which need to be maintained and updated
- Heterogeneity of software / hardware systems
- The pressure to deliver software quickly, at a low cost and with good quality

Development Models

- In order to develop a program, the following are necessary:
  - A clear understanding of what is required
  - A set of working methods and tools
  - An action plan

- Action plan template = model development

Software development process

- Requirements Engineering
- Arhitectural design
- Detailed design
- Implementation
- Integration
- Validation
- Verification
- Deployment
- Maintenance

Requirements Engineering

- It sets out what the customer wants the program to do
- The aim is to record the requirements in a clear and more accurate manner
- Problems
  - Communication
  - Negotiation
  - Advise client

Design

- Arhitectural
  - For reasons of complexity, large programs can not be designed and implemented as a single piece
  - The program is divided into modules or simpler components that can be individually addressed

- Detailed
  - Designing each module of the application, in the smallest details

Implementation + integration

- Implementation
  - Detailed design is translated into a programming language
  - This is achieved modularly, on the resulting structure of the architectural design

- Integration
  - Big-bang model
  - Incremental model

Validation and verification

- **Validation**: we ensure that the program meets the user requirements
  - *Are we building the correct product?*

- **Verification**: we ensure that the program is stable and working properly, in terms of development.
  - *Are we building the product correctly?*

Maintenance

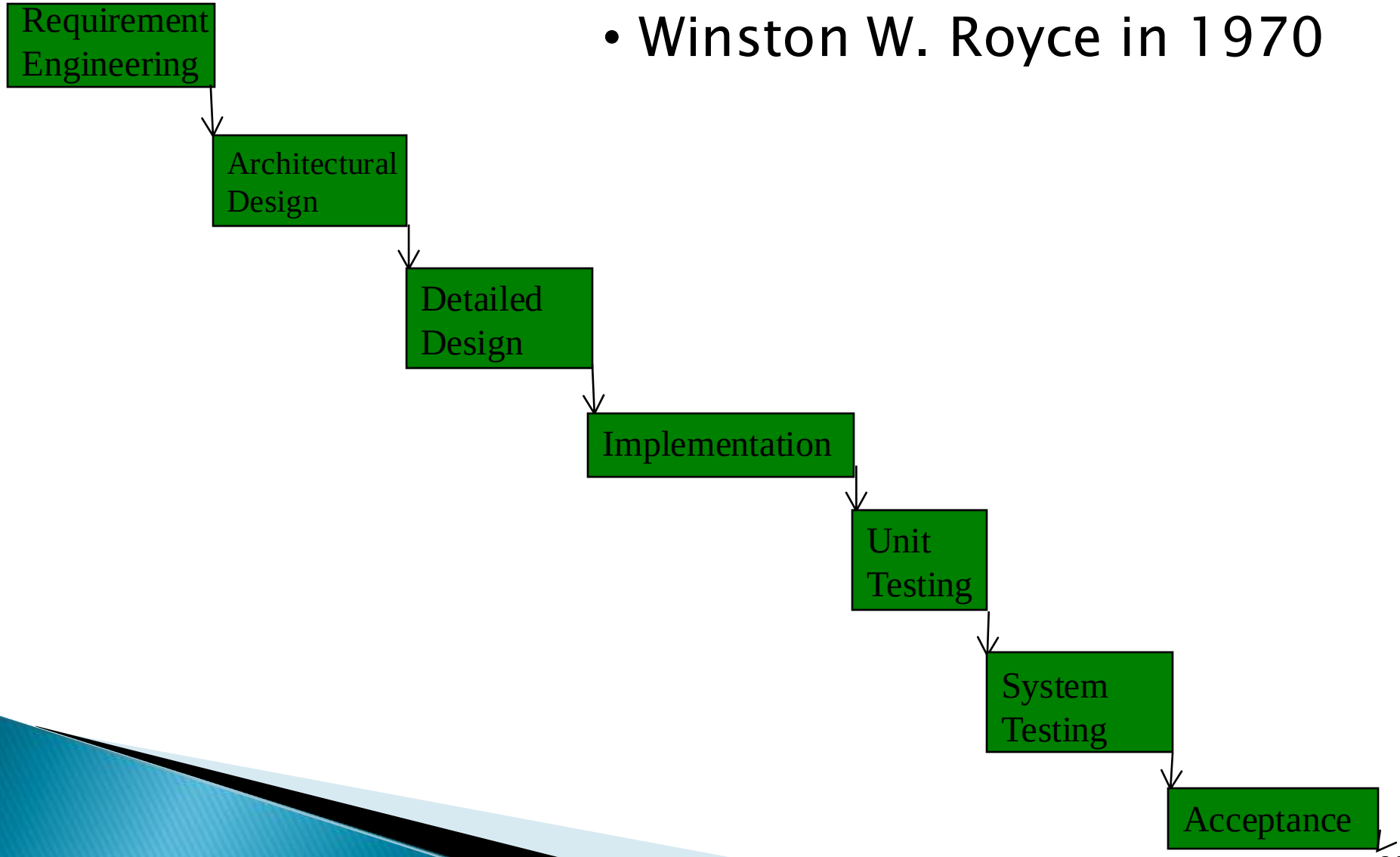
- After delivery
  - Discovered mistakes must be repaired
  - Changes in specifications may appear
  - There may be new requirements
  - The training of those who will use the product

- Maintenance = managing these types of problems

Development models

- *How to perform those activities specified by the stages of software development*
- *Examples of development models:*
  - *Ad hoc: manage as best you can*
  - *Waterfall (with feedback)*
  - *Prototyping*
  - *Spiral*
  - *RUP (Rational Unified Process)*
  - *V-Model*
  - *XP (Extreme Programming)*
  - *Agile, Lean, Scrum*
  - *MDD, AMDD*
  - *TDD*

Waterfall



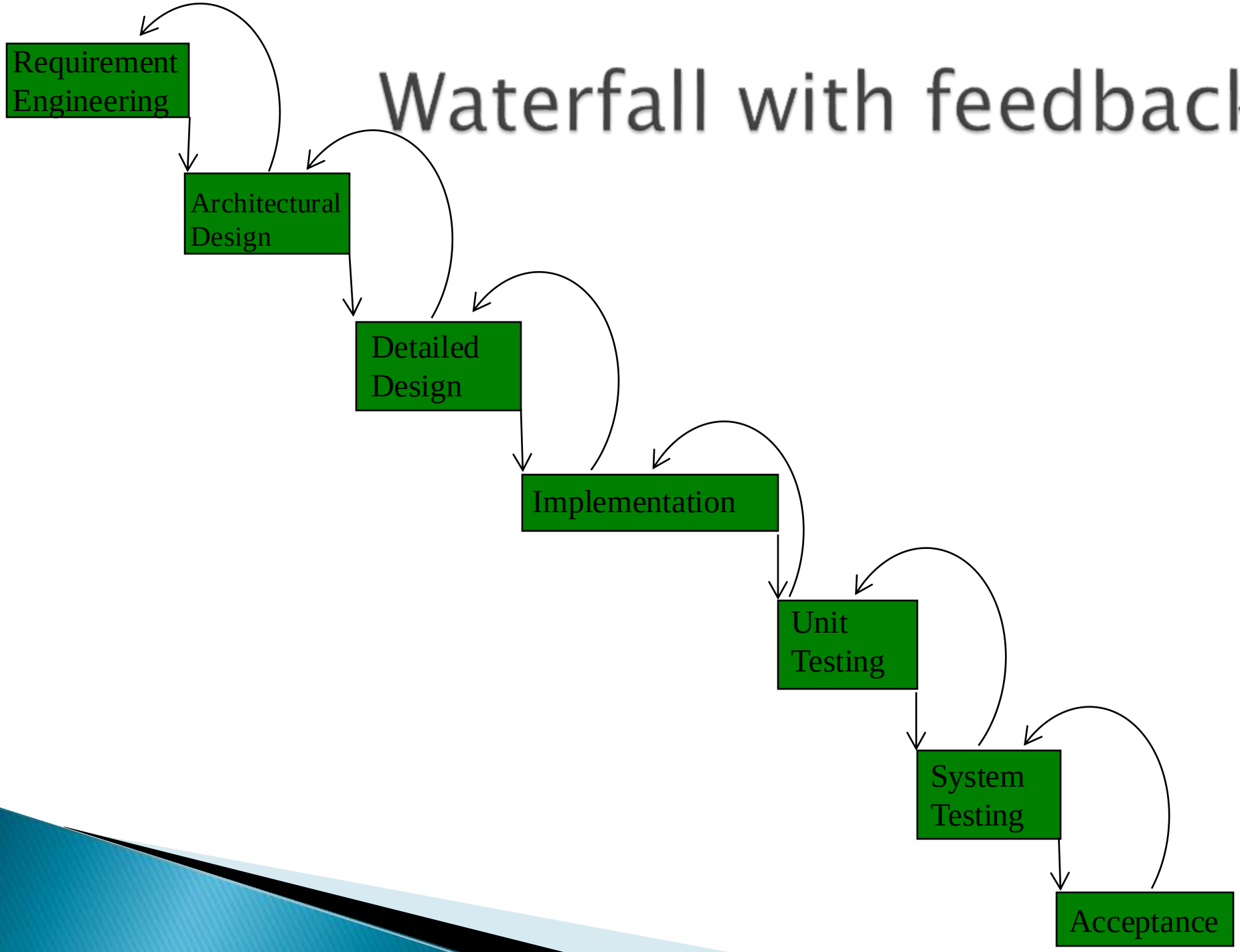
- Winston W. Royce in 1970

Waterfall+-

- +: Split a complex task into smaller steps
- +: Easy to manage and control
- +: Each step results in a well defined product
- +: Always know the phase of the development: what have we finished so far? What else do we have to do further?

- -: Errors are propagated between stages
- -: No repair mechanism for errors

Waterfall with feedback



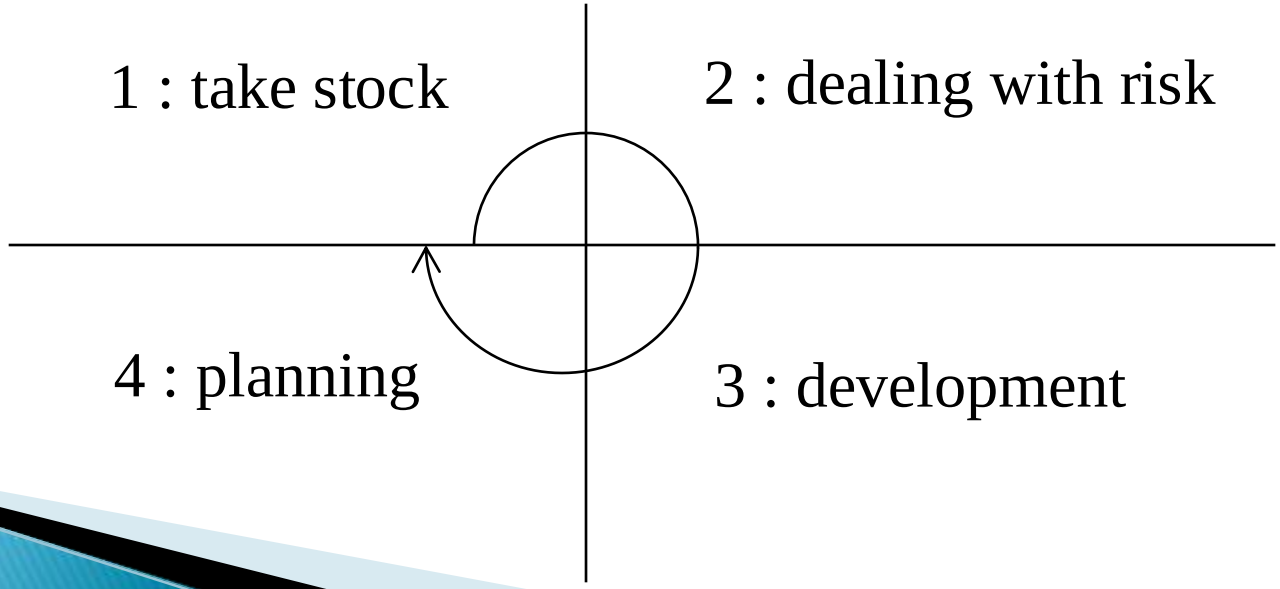
Waterfall with feedback+-

- +: Allows for fixing errors from the previous step
- -: Errors from step i that are discovered to step i + 2 are not remedied
- -: The customer sees the final product only at the end of development

Spiral

- Feasibility study
- Requirements analysis
- Architectural design
- Implementation

At each step, we do the following:



Spiral +-

- +: Keeps benefits from waterfall model
- +: Take into account the notion of risk

- Examples of risks:
  - A competing company launches a similar product
  - An architect leaves team
  - Changing in customer requirements
  - A team does not comply the delivery deadlines

Conclusions

- The statistics prove the importance of using engineering techniques in software development
- SE definitions use keywords such as: *engineering methods, large projects are implemented by teams, safe programs which work effectively, development, maintenance, planning, budgeting*
- Steps necessary for developing large projects
- Development Models: Ad-Hoc, Waterfall, Spiral

Questions

- Why is it useful to learn this subject?
- Have you encountered some of the elements of Software Engineering? Where?

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